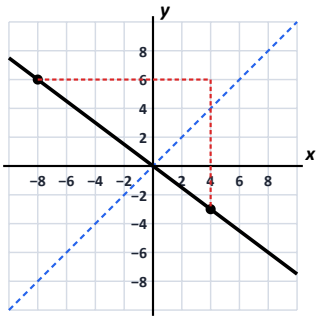
Exit Ticket Answer Key

Thursday, October 29 · Module 2, Topic 2, Lessons 1–3 · Form B (like the End of Topic Assessment) · question n matches End of Topic Assessment item n

1. f(x) = x, m(x) = a · f(x). What is a? **a = −3/4**

1. m(x) = a · x, so a is the slope of m.
2. From (−8, 6) go right 12 and down 9: slope = −9/12 = **−3/4**. Or (−3 − 6) / (4 − (−8)) = −9/12.
3. Check: m(4) = −3/4 · 4 = −3 ✓. f(x) = x is compressed by 3/4 and reflected over the x-axis.

3/4: missed the reflection (line falls). **−4/3:** did run over rise.



2. Range for $50 + $30 per hour, at most 8 hours. **C. 50 ≤ y ≤ 290**

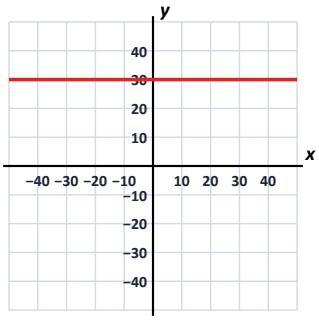
1. y = 30x + 50 with 0 ≤ x ≤ 8 (the domain).
2. x = 0 → y = 50. x = 8 → y = 30(8) + 50 = 290. Range: **50 ≤ y ≤ 290**.

B: that is the domain (x), not the range (y). **A:** started at the hourly rate instead of the fixed cost.

3. Equation and slope of the line. **H. y = 30; slope is zero**

1. Each grid line is 10. The line is **horizontal** through y = 30, so every point has y = 30.
2. Horizontal line: rise = 0, so the slope is **zero** (HOY). A vertical line x = c has an undefined slope.

F/J: mixed up x = and y =. **G:** mixed up zero (horizontal) and undefined (vertical).



4. g(x) = f(x + 4). **D. translated left 4 units**

1. +4 is **inside** the parentheses → inputs change → left/right. Inside works opposite the sign: + inside moves **left**.

A: treated +4 as outside (up). **C:** knew it was horizontal but went the way the sign looks.

5. Translate f(x) 3 units left. y-intercept? **(0, 5)**

1. Left 3: every x-value − 3, y-values stay. x = 0 on g came from x = 3 on f, and f(3) = 5.
2. Check: f(x) = 2x − 1, g(x) = f(x + 3) = 2(x + 3) − 1 = 2x + 5.

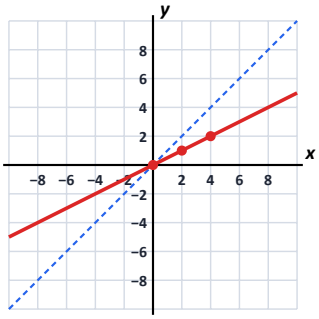
(0, −7): moved right (used f(−3)). **(0, 2):** moved up 3. **(0, −1):** the original y-intercept.

x	f(x)	x − 3
1	1	−2
2	3	−1
3	5	**0**

6. f(x) = x, g(x) = f(½x). **g(x) = ½x**

1. Replace x with ½x: g(x) = ½x. Plot (0, 0), (2, 1), (4, 2).
2. Horizontal stretch by 2: the line is less steep than f(x).

y = 2x (steeper): compressed instead of stretched. **y = x + ½:** added instead of multiplied.



7. Perpendicular to $y = \frac{1}{2}x - 3$, crosses the x-axis at −2. **$y = -2x - 4$**

1. Negative reciprocal of ½: flip → 2, change the sign → **−2**. Point: **(−2, 0)**.
2. $y - 0 = -2(x - (-2))$ → $y = -2(x + 2)$ → $y = -2x - 4$. Check: $-2(-2) - 4 = 0$ ✓

$y = -2x + 4$: sign error with $x - (-2)$. **slope 2 or −½:** only one step of flip-and-change-sign. **(0, −2):** used −2 as the y-intercept.

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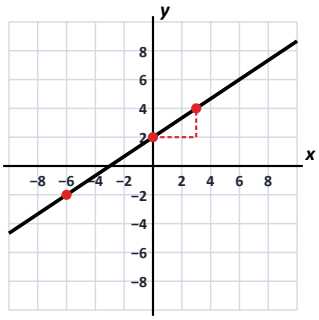
8. Parallel to $y = -2.5x + 1$, through (4, 0). **$y = -2.5x + 10$**

1. Parallel → same slope −2.5. $y - 0 = -2.5(x - 4)$ → $y = -2.5x + 10$. Check: $-2.5(4) + 10 = 0$ ✓

$-2.5x - 10$: sign error distributing. **$-2.5x + 1$:** kept the old y-intercept. **slope 0.4:** used the perpendicular slope.

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9. Point-slope form of the graph. **$y - 2 = \frac{2}{3}(x - 0)$**

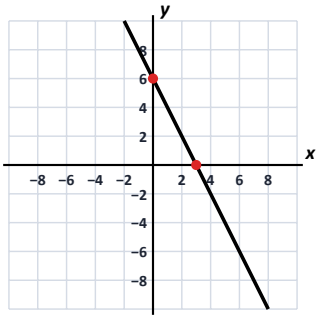


1. Find two grid points: (0, 2) and (3, 4). From (0, 2): right 3, up 2 → slope **2/3**.
2. $y - y_1 = m(x - x_1)$ → $y - 2 = \frac{2}{3}(x - 0)$.
3. Also correct: $y - 4 = \frac{2}{3}(x - 3)$, $y + 2 = \frac{2}{3}(x + 6)$, $y - 6 = \frac{2}{3}(x - 6)$. Any point on the line works.

slope −2/3: the line rises, so the slope is positive. **3/2:** run over rise. Sign errors: $y - (-2)$ is $y + 2$.

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10. Select TWO. **G and J**



1. The line falls left to right → decreasing (F is false). From (0, 6) right 1, down 2 → slope **−2**: **G** ✓.
2. Crosses the y-axis at (0, 6), so H is false. Crosses the x-axis at x = 3 → the zero is 3: **J** ✓ (K is the y-intercept value).
3. Equation: $y = -2x + 6$. Check: $-2(3) + 6 = 0$ ✓

K: mixed up the zero (x-axis) and the y-intercept. **H:** read the zero as the y-intercept.

Next step: Q1, 4, 5, 6 → Lesson 1 and 2 videos. Q7, 8 → Lesson 3 video and homework. Q2, 3, 9, 10 → earlier skills: review on Mon 11/2. Compare with Wednesday's Form A to see who improved.